# Statistical foundations of business analytics

## Day 1  Design of experiments

Metrics, funnels and evidence for a business decision

26 October 2026

Interactive lecture deck for a 150 minute session

# The lecture route

| Minutes | Topic | Notes pages |
|---|---|---|
| 0 to 10 | Business question and statistical recap | 2 |
| 10 to 30 | Metric design and guardrails | 3 to 4 |
| 30 to 45 | Funnels and their denominators | 5 to 6 |
| 45 to 70 | Causality and experimental design | 7 to 10 |
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Topic labels and answer links are clickable in the PDF.

# Should ShopEase shorten its checkout

The product team believes customers leave because checkout takes too much effort.

The proposed change reduces the number of checkout steps.

**Your decision: does the new experience create enough additional profitable purchasing to justify rollout?**

All customer data and business results in this deck are hypothetical.

# Purchases rose after launch

| Period | Eligible users | Purchasing users |
|---|---|---|
| Before | 1,000 | 80 |
| After | 1,000 | 100 |

## Would you credit the shorter checkout for the extra purchases?

Discuss for 30 seconds. Name two other explanations.

Show worked answer

# Several explanations fit these counts

A festival promotion could attract customers who already intend to buy.

Advertising, stock availability or the mix of customers could change.

Random variation can also change the observed number of purchasers.

**The experiment must separate the checkout change from these other explanations.**

Return to the question

# Counts conversion and relative improvement

| Quantity | Existing period | Later period |
|---|---|---|
| Purchasers / eligible users | 80 / 1,000 | 100 / 1,000 |
| Purchasing conversion | 8% | 10% |

$$10\% - 8\% = 2 \text{ percentage points}$$

$$\frac{10\% - 8\%}{8\%} = 25\% \text{ relative increase}$$

The baseline for the relative improvement is the original 8%.

# Conversion is the mean of zeros and ones

For each assigned customer, record 1 for a purchase within seven days and 0 for a fully observed nonpurchase.

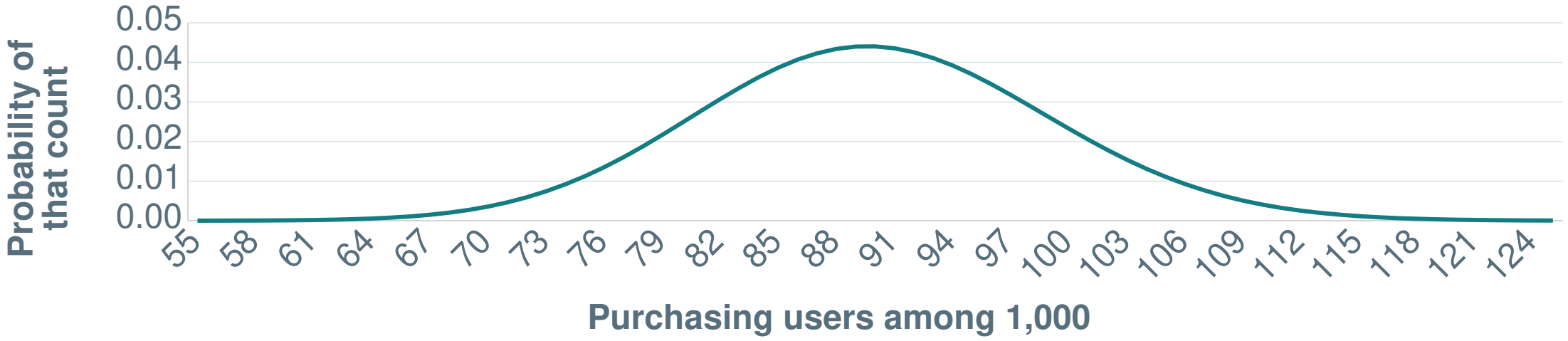
$$Y_i \in \{0, 1\}, \qquad \hat{p} = \frac{\sum_{i=1}^{n} Y_i}{n}$$

One purchasing customer among ten gives an average of 0.10, or 10%.

**A customer who places three orders still contributes one purchasing customer.**

# The observed rate varies across samples

Suppose the underlying purchase probability stays at 9% for comparable independent users.




p describes the underlying probability. The observed p-hat changes when the sample changes.

# A reproducible purchasing metric

Two analysts can report different conversion rates from the same data if they choose different units or windows.

**Our outcome: the fraction of eligible assigned accounts with at least one successful purchase within seven days.**

This sentence specifies who can enter, what counts as success and how long each person can purchase.

# One customer returns and buys twice

Meera enters checkout on Monday, leaves, and returns on her phone on Thursday.

She makes two successful purchases within her seven-day window.

**How much does she contribute to the numerator and denominator of purchasing conversion?**

Discuss with a neighbour before revealing the answer.

Show worked answer

# Meera contributes one to each count

| Part of the metric | Contribution |
| --- | --- |
| Eligible assigned accounts | 1 |
| Purchasing accounts within seven days | 1 |
| Purchase indicator for Meera | Y = 1 |

**Her two purchases can both contribute to revenue and margin. The binary conversion outcome counts her once.**

Return to the question

# Eligibility and assignment precede the outcome

Enroll a logged-in account at its first eligible checkout during the enrollment period.

Check the supported region and payment rules before assignment. Apply staff and test-account exclusions using fixed rules.

Randomly assign the account and retain that experience on later visits.

**Record one authoritative account identifier, assignment time and experiment arm.**

# Every customer receives a full seven days

| Account assigned | Its outcome window ends |
| --- | --- |
| 26 October at 10 a.m. | 2 November at 10 a.m. |
| Last enrollment day | Seven days after that assignment |

The clock starts at each account's assignment, rather than at the experiment's launch.

**If enrollment lasts 14 days, the final accounts still need their seven-day follow-up.**

# A zero and a missing outcome mean different things

| Situation | Interpretation |
| --- | --- |
| Complete reliable window, no purchase | Observed outcome 0 |
| Only one day has elapsed | Seven-day outcome is incomplete |
| Payment logging failed | Outcome may be missing |

A missing record does not establish a nonpurchase. The data pipeline and follow-up need investigation.

# Which denominator preserves the comparison

| Outcome | Existing checkout | Shorter checkout |
|---|---|---|
| Assigned accounts | 1,000 | 1,000 |
| Reach payment page | 500 | 800 |
| Purchase | 100 | 120 |

**Calculate conversion using all assigned accounts. Then calculate it using only payment-page visitors.**

Why do the apparent rankings differ?

Show worked answer

# Later behaviour changes the selected population

| Denominator | Existing checkout | Shorter checkout |
| --- | --- | --- |
| All assigned accounts | 10% | 12% |
| Payment-page visitors | 20% | 15% |

The shorter checkout may itself change who reaches payment. Conditioning on that later behaviour changes the groups being compared.

**The primary conversion denominator retains every eligible assigned account.**

Return to the question

# The metric definition another analyst receives

Seven-day purchasing conversion counts assigned eligible accounts with at least one successful purchase during the next seven 24-hour periods.

$$\text{Conversion} = \frac{\text{assigned accounts purchasing within seven days}}{\text{all eligible assigned accounts}}$$

Use a consistent payment definition, account identity, time zone and window boundary.

**Retain the original assignment across visits and count each account once.**

# Primary diagnostic and guardrail metrics

| Role | Example |
| --- | --- |
| Primary | Purchasing conversion per assigned account |
| Diagnostic | Checkout step completion |
| Business guardrail | Contribution per assigned account |
| Reliability guardrail | Users with payment failure per assigned account |

Choose these roles and acceptable guardrail changes before looking at the results.

# More purchasers can change contribution

A separate pricing example. The contribution amounts include the complete contribution per purchaser during the window.

| Quantity | Regular price | Discount |
|---|---|---|
| Assigned accounts | 1,000 | 1,000 |
| Purchasing accounts | 80 | 100 |
| Contribution per buyer | INR 300 | INR 200 |

## Which arm has higher contribution per assigned account?

Show worked answer

# Observed contribution falls despite higher conversion

| Quantity | Regular price | Discount |
|---|---|---|
| Conversion | 8% | 10% |
| Total contribution | INR 24,000 | INR 20,000 |
| Contribution / assigned account | INR 24 | INR 20 |

Contribution uses an agreed revenue-minus-variable-cost definition.

**These are observed means. A population margin claim also needs an uncertainty analysis.**

Return to the question

# Two channels need an explicit weighting rule

| Channel | Users | Buyers | Conversion |
|---|---|---|---|
| A | 100 | 20 | 20% |
| B | 900 | 90 | 10% |

## Is conversion across these 1,000 users 15% or 11%?

Explain what each calculation weights equally.

Show worked answer

# Pooled customer conversion is 11 percent

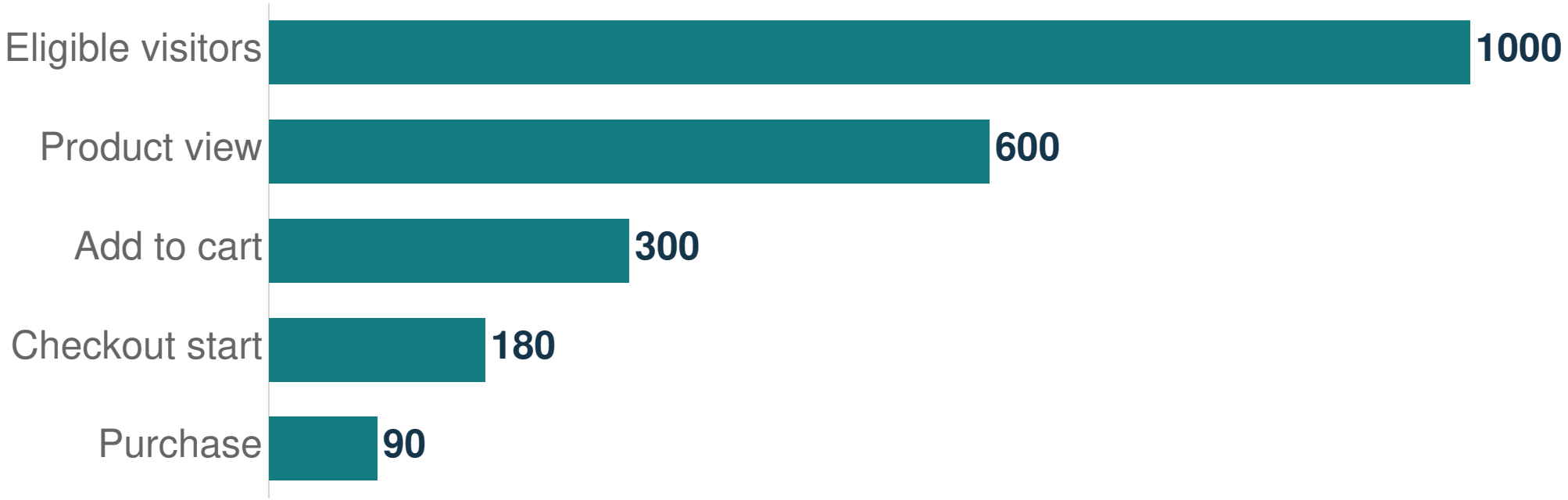
$$\frac{20 + 90}{100 + 900} = \frac{110}{1000} = 11\%$$

The 15% average gives each channel equal weight, although channel B has nine times as many users.

**Use weights that match the intended population and business question.**

Return to the question

# A funnel follows an ordered customer journey




Each later group is a subset of the preceding group within the same window.

# Stage conversion and overall conversion

| Stage | Unique users | From previous stage | From entry |
|---|---|---|---|
| Entry | 1,000 | Not applicable | 100% |
| Product view | 600 | 60% | 60% |
| Add to cart | 300 | 50% | 30% |
| Checkout start | 180 | 60% | 18% |
| Purchase | 90 | 50% | 9% |

90 / 180 describes checkout completion. 90 / 1,000 describes overall visitor conversion.

# Nested denominators explain the product rule

$$\frac{600}{1000} \times \frac{300}{600} \times \frac{180}{300} \times \frac{90}{180} = \frac{90}{1000} = 0.09$$

The intermediate counts cancel because the same ordered cohort supplies each numerator and denominator.

**This identity requires consistent nesting, identity and observation windows.**

Unrelated dashboard percentages do not automatically satisfy these conditions.

# What if checkout completion reached 60 percent

The current funnel has 1,000 entrants, 180 checkout starters and 90 purchasers.

Suppose checkout completion rises from 50% to 60%, while every earlier stage rate stays fixed.

**How many purchases does this planning calculation predict? What assumption carries the result?**

Show worked answer

# The planning count becomes 108 purchases

$$180 \times 0.60 = 108, \qquad \frac{108}{1000} = 10.8\%$$

That is 18 additional purchasers per 1,000 entrants, or 1.8 percentage points.

The calculation assumes the earlier stage rates, population and measurement stay unchanged.

**An experiment is still needed to establish whether the proposed change produces that improvement.**

Return to the question

# Raw event rows and analysis units differ

| Source table | Typical records |
| --- | --- |
| Assignment | One persistent assignment per account |
| Events | Many visits and events per account |
| Orders and payments | Several purchases or line items per account |

Build one binary outcome row per assigned account for the conversion analysis.

**Aggregate order line items before joining so the same revenue does not appear repeatedly.**

# A funnel drop identifies an investigation

Browsing, shipping charges, payment failures and price comparisons can all produce abandonment.

A 9% visitor conversion and a higher conversion among checkout entrants can both be correct. Their entry populations differ.

**For 1,000 users, 4,000 visits and 90 buyers, user conversion is 90 / 1,000 = 9%.**

Checks include duplicate events, persistent assignment, successful payments and complete follow-up.

# The causal question includes an unseen alternative

For one customer, imagine the purchase outcome under the old checkout and the outcome under the shorter checkout.

The experiment observes only the version that customer receives. The other outcome is the counterfactual.

$$\tau = E(Y_i(1) - Y_i(0))$$

**Random assignment makes enrolled groups comparable in expectation before the intervention.**

# Random sampling and random assignment

| Question | Random process | Purpose |
|---|---|---|
| Who enters the study | Sampling from a population | Supports generalization |
| Who receives each version | Assignment within the study | Supports causal comparison |

A randomized trial among logged-in desktop users does not automatically establish the same effect for anonymous mobile visitors.

**The target population remains part of the claim.**

# Three comparisons have different weaknesses

| Comparison | Main concern |
| --- | --- |
| Last week versus this week | Calendar conditions and changing visitors |
| People who choose a feature versus others | Self-selection and motivation |
| Concurrent randomized groups | Random variation and implementation validity |

Randomization does not repair broken measurement, delivery errors or interference.

# Experimental language for the checkout case

| Term | Meaning |
| --- | --- |
| Factor and levels | Checkout design with existing and shorter levels |
| Response | Seven-day purchase indicator |
| Experimental unit | The account independently assigned |
| Control | The reference checkout |
| Confounding | Effects the comparison cannot separate |

# Four visits from one account share one person

A trial assigns 1,000 accounts and observes 4,000 visits.

Visits from the same account share buying intent, history and treatment assignment.

**Treating them as 4,000 independent user replications can exaggerate precision.**

For this conversion metric, retain one outcome per independently assigned account.

# A store experiment observes 10,000 shoppers

The company assigns 12 stores to two layouts, with six stores per layout.

It then records outcomes from 10,000 shoppers.

**How many units received independent treatment assignments?**
**Can we use the simple independent-user formulas?**

Show worked answer

# The experiment randomized 12 stores

**The store is the experimental unit. Shoppers within the same store share a treatment and operating conditions.**

Their observations provide outcome information, but do not create 10,000 independently assigned stores.

The analysis must account for clustering. The independent-account formulas in this lecture need modification.

Return to the question

# Randomization replication and blocking

Randomization assigns treatment by chance and prevents deliberate allocation by customer quality.

Replication supplies enough independently assigned units to quantify random variation.

Blocking randomizes within relevant groups known before treatment, such as device type.

**These principles improve the comparison while preserving a clear experimental unit.**

# Both versions appear within each device block

| Pre-treatment block | Existing checkout | Shorter checkout |
|---|---|---|
| Mobile | 50% of mobile accounts | 50% of mobile accounts |
| Desktop | 50% of desktop accounts | 50% of desktop accounts |

Giving the shorter checkout to every mobile user and the old checkout to every desktop user confounds device with treatment.

**Weight block-specific effects for the intended customer mix.**

# A protocol fixes the question before launch

State the business decision and why the proposed intervention might change purchasing.

Specify eligibility, account identity, control, treatment, allocation and any blocks.

Define the primary metric, guardrails, acceptable changes and observation windows.

**Choose an effect worth detecting and plan the required sample.**

# The protocol also fixes analysis and follow-up

Prespecify the test, significance level, enrollment target and stopping rule.

Allow every assigned account to finish its outcome window.

Check treatment rendering, payment joins, persistent assignment and event logging.

**Assess validity, effect uncertainty, business value and guardrail evidence together.**

# Allocation and delivery deserve their own checks

An A/A test assigns two identical experiences to check parts of the implementation and analysis pipeline.

Small departures from the intended 50:50 allocation are expected. An unusually large departure can signal a sample ratio mismatch.

Investigate assignment, filtering and logging before interpreting the effect.

**Retain original assignments in the primary assignment-based analysis.**

# The manager reserves the new checkout for loyal users

Returning customers receive the shorter checkout. New customers receive the existing checkout.

The shorter checkout has higher conversion and a very small p-value.

**What prevents a causal conclusion? How would you repair the comparison?**

Show worked answer

# Customer familiarity is entangled with checkout

Returning customers may already trust the store and know how to buy.

The comparison combines their preexisting familiarity with the checkout design.

**Randomize comparable eligible accounts concurrently between both versions.**

A smaller p-value cannot fix the assignment rule.

Return to the question

# Five minute break

## Resume at minute 75.

Next: ShopEase wants to test a discount and free shipping together.

Think about whether one offer could change the effect of the other.

# Two factors create four treatment conditions

A separate promotional experiment varies a discount D and free shipping F. 0 means no; 1 means yes.

| Discount D | Free shipping F | Users | Buyers | Conversion |
|---|---|---|---|---|
| 0 | 0 | 1,000 | 80 | 8% |
| 1 | 0 | 1,000 | 100 | 10% |
| 0 | 1 | 1,000 | 110 | 11% |
| 1 | 1 | 1,000 | 180 | 18% |

Each condition needs independently assigned users. Run the four conditions concurrently.

# Does the discount have one effect

| Shipping setting | No discount | Discount |
|---|---|---|
| Paid shipping | 8% | 10% |
| Free shipping | 11% | 18% |

**Calculate the discount effect separately under paid and free shipping.**

Which setting produces the larger observed effect?

Show worked answer

# The conditional discount effects are 2 and 7 points

$$\text{Paid shipping: } 10\% - 8\% = 2 \text{ points}$$

$$\text{Free shipping: } 18\% - 11\% = 7 \text{ points}$$

**The discount effect depends on the shipping setting on this additive probability scale.**

These are observed contrasts. Population claims still require uncertainty analysis.

Return to the question

# Main effects average over the other factor

$$\text{Discount: } \frac{10+18}{2} - \frac{8+11}{2} = 4.5 \text{ points}$$

$$\text{Shipping: } \frac{11+18}{2} - \frac{8+10}{2} = 5.5 \text{ points}$$

These averages give the two settings of the other factor equal weight.

**A 4.5-point discount main effect differs from the 2-point effect under an always-paid-shipping policy.**

# Interaction is a difference between effects

$$(18 - 11) - (10 - 8) = 7 - 2 = 5 \text{ percentage points}$$

An interaction occurs when the effect of one factor changes with the level of the other.

**Here the scale is a difference in purchase probabilities, expressed in percentage points.**

Testing the discount only with paid shipping would miss the larger observed discount effect with free shipping.

# The shipping lines have different slopes




The steeper line corresponds to a larger conditional discount effect.

# An additive prediction would give 13 percent

$$8\% + (10\% - 8\%) + (11\% - 8\%) = 13\%$$

The observed combined cell is 18%. It lies five percentage points above this no-interaction additive calculation.

**The 5-point excess matches the interaction contrast.**

This describes the chosen probability scale and these observed cell means.

# Would you deploy both offers

Discount plus free shipping has the largest observed conversion, 18%.

## Would you recommend rollout now?

Name the evidence you need about business value and uncertainty.

Show worked answer

# Offer costs remain part of the decision

Request contribution per assigned account, fulfillment and discount costs, reliability outcomes and uncertainty.

**A conversion interaction does not automatically imply a profitability interaction.**

The strongest observed cell is a candidate policy whose business performance still needs evaluation.

Return to the question

# A randomized checkout trial now uses the same counts

From here, assume a valid concurrent randomized experiment with independent accounts and complete seven-day follow-up.

| Arm | Assigned accounts | Purchasing accounts | Conversion |
|---|---|---|---|
| A existing checkout | 1,000 | 80 | 8% |
| B shorter checkout | 1,000 | 100 | 10% |

$$\hat{\Delta} = \hat{p}_B - \hat{p}_A = 0.10 - 0.08 = 0.02$$

How precisely does this estimate the population effect?

# Binary outcomes have variance p times one minus p

$$E(Y_i) = p, \qquad Y_i^2 = Y_i$$

$$\mathrm{Var}(Y_i) = E(Y_i^2) - E(Y_i)^2 = p - p^2 = p(1-p)$$

A customer outcome is either zero or one. The variance describes variation across such outcomes.

The uncertainty in an average conversion rate is smaller than this individual-level variation.

# Independent replication reduces uncertainty in the mean

$$\mathrm{Var}(\hat{p}) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$$

$$\widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Standard error describes the spread of the estimated conversion across repeated comparable samples.

These formulas assume independent user outcomes. Repeated visits and clustered assignments need suitable methods.

# Uncertainty from the two arms adds

$$\widehat{SE}(\hat{\Delta}) = \sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{n_A} + \frac{\hat{p}_B(1-\hat{p}_B)}{n_B}}$$

$$\sqrt{\frac{0.08(0.92)}{1000} + \frac{0.10(0.90)}{1000}} = 0.01279$$

The standard error is 0.01279 in proportions, or about 1.279 percentage points.

# The confidence interval includes uncertainty

$$\hat{\Delta} \pm 1.96\widehat{SE}(\hat{\Delta}) = 0.02 \pm 0.02507$$

| Scale | Lower limit | Estimate | Upper limit |
|---|---|---|---|
| Proportion | -0.00507 | 0.02000 | 0.04507 |
| Percentage points | -0.51 | 2.00 | 4.51 |

This approximate Wald interval contains a small negative effect and useful positive effects.

# A positive estimate can remain inconclusive

The observed effect is positive, but the interval includes zero.

**This analysis does not establish a positive population effect at the conventional two-sided 5% level.**

Across repeated comparable experiments, the interval procedure aims for 95% coverage of the fixed population effect.

It does not describe 95% of customers or a posterior probability for this realized interval.

# The same rates can have different precision

| Users per arm | Estimated difference | Approximate 95% interval |
|---|---|---|
| 1,000 | 2 percentage points | -0.51 to 4.51 points |
| 10,000 | 2 percentage points | 1.21 to 2.79 points |

The second row is a separate hypothetical dataset with the same 8% and 10% rates.

**More independent observations narrow the interval without changing the estimated difference.**

# The equality test starts from a no-effect model

$$H_0 : p_B - p_A = 0, \qquad H_1 : p_B - p_A \neq 0$$

Under the null, both arms share one purchasing probability. Estimate that common probability by pooling.

$$\hat{p}_{\text{pool}} = \frac{80 + 100}{1000 + 1000} = 0.09$$

The test and the interval use different standard errors because they answer different calculation questions.

# The observed difference is about 1.56 null standard errors

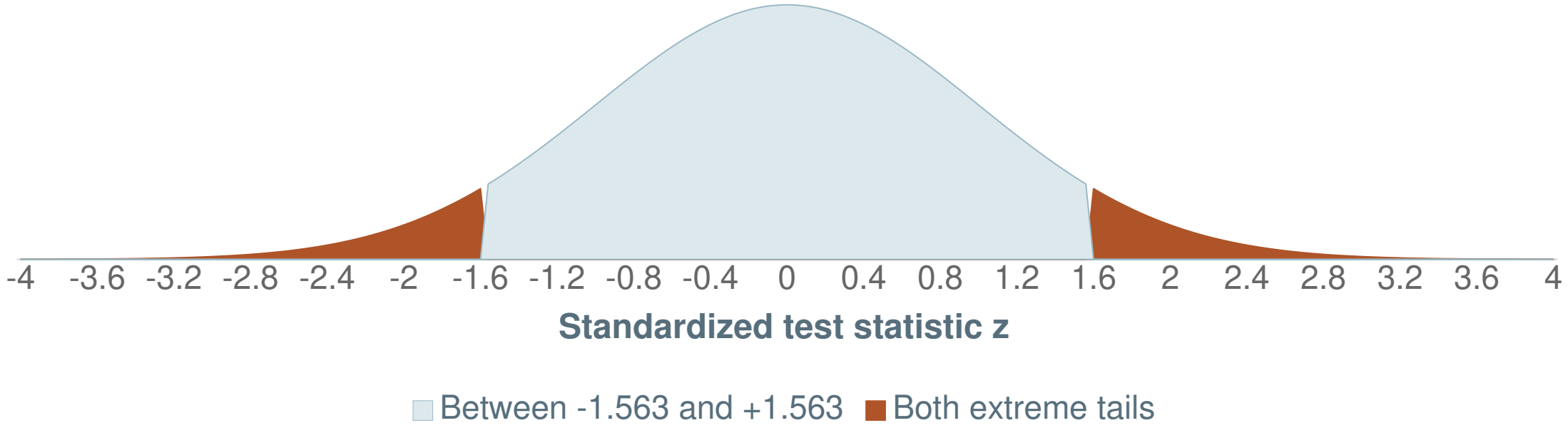
$$SE_0 = \sqrt{0.09(0.91)\left(\frac{1}{1000} + \frac{1}{1000}\right)} = 0.01280$$

$$z = \frac{0.10 - 0.08}{0.01280} = 1.563$$

**The two-sided p-value is approximately 0.118.**

Use the normal approximation when the design assumptions and success/failure counts are adequate.

# The p value includes both tails of the null model




Under the null model, about 11.8% of repeated test statistics would be at least this extreme in magnitude.

# What does p equal to 0.118 tell us

A colleague says: There is an 11.8% probability that the null hypothesis is true.

Another says: The experiment proves the checkout has no effect.

**How would you correct both statements?**

Show worked answer

# The analysis remains inconclusive about a positive effect

The p-value concerns extreme data under the specified null model and test assumptions.

At the planned 5% level, we fail to reject equality. This does not establish a zero effect.

**The interval still contains effects that could matter to the business.**

Choose a one-sided direction before seeing results when the scientific and business question justifies it.

Return to the question

# Errors and power belong in experiment planning

| Situation | Statistical consequence |
| --- | --- |
| Reject a true equality null | Type I error |
| Fail to reject at a particular real effect | Type II error |
| Reject at that specified real effect | Power = 1 minus Type II error probability |

A planned significance level of 5% controls false rejection under its assumptions. It does not mean 5% of reported positives are false.

# The minimum detectable effect is a planning choice

| Planning input | Example |
| --- | --- |
| Baseline and alternative conversion | 8% and 10% |
| Effect to detect | 2 percentage points |
| Significance and power | 5% two-sided and 80% |
| Approximate equal-arm requirement | 3,213 independent accounts per arm |

An MDE is an effect size used for a chosen power. Detection is not guaranteed for every effect above it.

# Enrollment and follow-up require different time

About 6,426 total accounts at 500 eligible accounts per day require roughly 13 enrollment days.

**A 14-day enrollment plan plus seven days for the last accounts gives about 21 elapsed days.**

$$SE \propto \frac{1}{\sqrt{n}} \qquad \text{Half the SE requires about four times } n$$

Traffic variation, logging delays and other metric windows also affect the calendar.

# A rollout decision needs the effect and its uncertainty

In the valid checkout trial, conversion is 8% versus 10%. The estimate is +2 points with an approximate 95% interval from -0.51 to +4.51 points.

For 100,000 similar future users, the point estimate suggests 2,000 extra purchasing users.

At an assumed INR 250 contribution each, that projects INR 500,000 before implementation and other incremental costs.

The projection inherits uncertainty in the effect, generalization and contribution assumptions.

# A fixed test needs a fixed analysis plan

Stopping when an ordinary fixed-sample p-value first falls below 0.05 changes its false-positive behaviour.

Use the planned analysis or a valid prespecified sequential method.

Searching many outcomes and segments also creates multiple-comparison problems.

**Distinguish the prespecified primary claim from exploratory findings that need appropriate adjustment or confirmation.**

# A nonsignificant guardrail result does not establish safety

Suppose the policy tolerates at most INR 0.50 loss in contribution per assigned user.

A wide interval can contain both acceptable and unacceptable losses.

## Assess the agreed tolerance explicitly and report uncertainty.

A confidence interval containing zero also does not establish equivalence. That needs a prespecified equivalence range and suitable analysis.

# Pair exercise for 25 minutes

| Task | Suggested time |
|---|---|
| 1 Conversion funnel | 4 minutes |
| 2 Randomized promotional offer | 8 minutes |
| 3 Factorial contrasts | 5 minutes |
| 4 Repair a proposed experiment | 3 minutes |
| Debrief using worked answers | 5 minutes |

Use each question's answer link during review. Submit your calculations and a short business recommendation.

# Exercise 1 on a conversion funnel

| Stage | Users |
|---|---|
| Entry | 1,200 |
| Cart | 600 |
| Checkout | 360 |
| Purchase | 144 |

Find stage conversions and overall conversion. Verify the product.

**If checkout completion rises to 50%, predict purchases and state the assumption. Write the overall metric definition.**

Show worked answer

# Exercise 2 on a randomized offer

| Arm | Assigned | Buyers | Contribution / buyer |
|---|---|---|---|
| A regular | 2,000 | 160 | INR 300 |
| B promotion | 2,000 | 220 | INR 200 |

Calculate conversion, absolute and relative differences, standard error and the approximate 95% interval.

Use p = 0.00121 for the planned two-sided test. Calculate total contribution and contribution per assigned user.

**Checks pass and follow-up is complete. Recommend rollout?**

Show worked answer

# Exercise 3 on two promotional factors

| Discount D | Delivery benefit F | Users | Buyers |
|---|---|---|---|
| 0 | 0 | 1,000 | 100 |
| 1 | 0 | 1,000 | 120 |
| 0 | 1 | 1,000 | 130 |
| 1 | 1 | 1,000 | 200 |

Find discount effects within each delivery setting, both equally weighted main effects, and the interaction.

**Explain why the largest conversion cell does not establish the best business policy.**

Show worked answer

# Exercise 4 on repairing the experiment

The manager proposes:

Use the old checkout this week and the new checkout next week. Count every visit as independent. Remove abandoners. Check p every morning and stop when p is below 0.05.

**Identify four problems and a correction for each. Write a two-sentence replacement plan.**

Show worked answer

# Exercise 1 worked answer

$$\frac{600}{1200} = 50\%, \quad \frac{360}{600} = 60\%, \quad \frac{144}{360} = 40\%$$

$$0.50 \times 0.60 \times 0.40 = 0.12 = 12\%$$

At 50% checkout completion: 180 purchases; overall conversion = 15%.

Assume earlier stage rates, population and measurement stay fixed.

**Metric: eligible unique entrants purchasing in a fixed common window / all eligible unique entrants.**

Return to the question

# Exercise 2 conversion and uncertainty

$$\hat{p}_A = 0.08, \quad \hat{p}_B = 0.11, \quad \hat{\Delta} = 0.03$$

$$\text{Relative increase} = \frac{0.03}{0.08} = 37.5\%$$

$$SE = 0.009260, \quad 95\%\ \text{CI: } 1.19 \text{ to } 4.81 \text{ points}$$

**p = 0.00121 is below 0.05. Under the stated valid design, the planned analysis supports increased conversion.**

Return to the question

# Exercise 2 contribution and the recommendation

| Quantity | A regular | B promotion |
|---|---|---|
| Total contribution | INR 48,000 | INR 44,000 |
| Contribution / assigned user | INR 24 | INR 22 |

Observed contribution per assigned user falls by INR 2. Conversion alone does not justify broad rollout.

**Evaluate the contribution effect and its uncertainty against the agreed business tolerance. User-level variation matters.**

Return to the question

# Exercise 3 factorial contrasts

| Quantity | Calculation | Result |
|---|---|---|
| Discount with no benefit | 12 - 10 | 2 points |
| Discount with benefit | 20 - 13 | 7 points |
| Discount main effect | (2 + 7) / 2 | 4.5 points |
| Delivery main effect | [(13 - 10) + (20 - 12)] / 2 | 5.5 points |
| Interaction | 7 - 2 | 5 points |

The largest conversion cell can carry the largest offer costs. Profitability and uncertainty still matter.

Return to the question

# Exercise 4 repairs

| Problem | Repair |
| --- | --- |
| Consecutive weeks | Randomize concurrently |
| Visits treated as independent users | Assign accounts and derive account outcomes |
| Abandoners removed | Retain all eligible assigned accounts |
| Stop on first small ordinary p-value | Use a fixed plan or valid sequential method |

Randomize accounts concurrently and retain every assignment. Analyze planned account outcomes after full follow-up using a fixed sample or valid sequential plan.

Return to the question

# Five exit questions

1 What are the absolute and relative changes from 8% to 10%?

2 Why are four visits from one account not four independent user replications?

3 What can an interval containing zero establish?

4 When do two factors interact?

5 Does a nonsignificant guardrail test demonstrate acceptability?

Show worked answer

# The answers that support a defensible decision

1 The change is 2 percentage points, or 25% relative to 8%.

2 Those visits share one person and treatment assignment.

3 The analysis has not established a nonzero effect. Equivalence needs its own criterion.

4 One factor's effect depends on the other's level on the specified scale.

**5 Guardrails need evidence against their prespecified tolerance.**

Return to the question

# Optional spreadsheet inputs and interval

Enter A in row 2 and B in row 3. Column B contains assigned counts and column C contains purchasing counts.

| Cell | Value or formula |
|---|---|
| B2 and C2 | 1000 and 80 |
| B3 and C3 | 1000 and 100 |
| D2 and D3 | =C2/B2 and =C3/B3 |
| B6 | =D3-D2 |
| B8 | =SQRT(D2*(1-D2)/B2+D3*(1-D3)/B3) |
| B9 and B10 | =B6-1.96*B8 and =B6+1.96*B8 |

# Optional spreadsheet test calculation

| Cell | Formula |
| --- | --- |
| B12 pooled conversion | =(C2+C3)/(B2+B3) |
| B13 null standard error | =SQRT(B12*(1-B12)*(1/B2+1/B3)) |
| B14 statistic | =B6/B13 |
| B15 two-sided p-value | =2*(1-NORM.S.DIST(ABS(B14),TRUE)) |

Use 0.08 in formulas for 8%. Multiply a difference in proportions by 100 to report percentage points.

# Optional sample size calculation

$$n \approx \frac{\left[z_{1-\alpha/2}\sqrt{2\bar{p}(1-\bar{p})} + z_{1-\beta}\sqrt{p_0(1-p_0)+p_1(1-p_1)}\right]^2}{(p_1-p_0)^2}$$

p0 = 0.08, p1 = 0.10, average p = 0.09, z0.975 = 1.95996 and z0.80 = 0.84162.

**The approximation gives 3,212.94, rounded up to 3,213 per arm.**

The formula omits continuity, clustering, multiplicity and sequential adjustments. Plan for the actual design.

# Optional contribution uncertainty

For the offer worksheet only, assume every buyer contributes exactly INR 300 in A and INR 200 in B. Nonbuyers contribute zero.

$$SE = \sqrt{\frac{300^2(0.08)(0.92)}{2000} + \frac{200^2(0.11)(0.89)}{2000}} = 2.296$$

**The estimated difference is -INR 2 per assigned user. The approximate 95% interval is -INR 6.50 to +INR 2.50.**

This does not demonstrate acceptability for a -INR 0.50 loss tolerance. Variable margins require their observed user-level variance.

# Optional equation for the four observed cell means

$$\hat{p}(D, F) = 0.08 + 0.02D + 0.03F + 0.05DF$$

D and F take values zero or one. The four substitutions reproduce 8%, 10%, 11% and 18%.

**Under this coding, the 0.05 coefficient equals the five-percentage-point interaction contrast.**

The equation represents these cell means. Other coding conventions scale coefficients differently.

# Important interpretation boundaries

A statistically significant tiny effect can have little business value.

A wide interval containing zero can contain both important benefits and important harms.

Selecting only users who actually engage with a treatment can break the randomized comparison.

**Higher average order value does not by itself establish higher revenue per assigned user.**

# Source notes and design references

Source: Day 1 Detailed Teaching Notes, 25 pages. Page references on the slides connect the deck to the original guide.

NIST full factorial designs
https://www.itl.nist.gov/div898/handbook/pri/section3/pri333.htm

NIST randomized block designs
https://www.itl.nist.gov/div898/handbook/pri/section3/pri332.htm

Microsoft Research, Patterns of Trustworthy Experimentation, Pre Experiment Stage
https://www.microsoft.com/en-us/research/articles/patterns-of-trustworthy-experimentation-pre-experiment-stage/

# Inference and implementation references

NIST difference of proportions confidence limits
https://www.itl.nist.gov/div898/software/dataplot/refman1/auxillar/diffprop.htm

OpenStax Introductory Business Statistics 2e, section 10.4
https://openstax.org/books/introductory-business-statistics-2e/pages/10-4-comparing-two-independent-population-proportions

Microsoft Research, Diagnosing Sample Ratio Mismatch in A/B Testing
https://www.microsoft.com/en-us/research/articles/diagnosing-sample-ratio-mismatch-in-a-b-testing/

All cases and numerical exercises are hypothetical teaching examples.